GNSS for Hurricane Monitoring: Tracking Storm Surge and Varying Atmosphere

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ABSTRACT

Hurricane Ian (September 23–30, 2022) was among the most destructive hurricanes to impact the southeastern United States, generating severe storm surge, flooding, and widespread coastal damage. Monitoring sea level and precipitable water vapor during such extreme events is critical for assessing coastal vulnerability and strengthening hazard resilience. In this study, Global Navigation Satellite System Interferometric Reflectometry (GNSS-IR), GNSS-derived tropospheric delays (GNSS-Trop), and tide gauge records were used to track sea level variations and water vapor conditions along Hurricane Ian's path.

GNSS-IR sea level estimates complement tide gauge observations, improving the compatibility of coastal storm-driven anomaly monitoring. In parallel, GNSS-Trop results were examined to characterize atmospheric water vapor changes, with GNSS-derived Integrated Water Vapor (IWV) compared against satellite-based precipitation estimates (GPM/IMERG). These results highlighted the links between the spatial and temporal patterns of water vapor variability, storm intensity, and landfall.

Overall, this study demonstrates the utility of terrestrial GNSS as a complementary tool to tide gauges and meteorological instruments, with integrated observations offering a robust framework for advancing coastal hazard monitoring.

DATASET

This study focuses on Hurricane Ian across the Southeastern United States, Caribbean, Puerto Rico, and the U.S. Virgin Islands during September 23–30, 2022, corresponding to DOY 266–273. Several GNSS stations were analyzed across the storm region; however, the poster highlights SCFJ, N003, and N300 because these stations showed clearer storm-related responses during Hurricane Ian.

The dataset integrates multiple observation sources to capture both coastal water-level response and atmospheric moisture variability. GNSS-IR data were obtained from GNSSrefl RINEX/SNR files and the SONEL database for sea-level anomaly estimation. Tide gauge records from NOAA were used to validate GNSS-derived sea-level changes. GNSS-Trop products from NGL were used to extract integrated water vapor, while GPM/IMERG precipitation data from NASA Giovanni were used to characterize rainfall conditions during the event.

Together, these datasets support analysis of storm-surge response, atmospheric water vapor evolution, and precipitation patterns along Hurricane Ian's track.

Table 1. Datasets Used for Hurricane Ian Coastal and Atmospheric Analysis

Dataset	Purpose
GNSS-IR	Sea Level anomaly
GNSS-Trop	Integrated Water Vapor
Tide Gauge	Validation
GPM/IMERG Precipitation	Precipitation

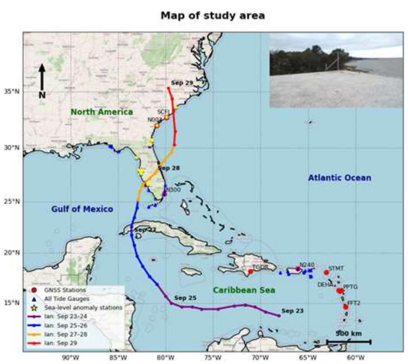


Figure 1. Study Area and Hurricane Ian Track. Map showing the study region across the Southeastern United States, Caribbean, Puerto Rico, and U.S. Virgin Islands, including Hurricane Ian's track from September 23–29, 2022, GNSS station locations, tide gauges, and sea-level anomaly stations.

METHODOLOGY

GNSS-IR analysis was performed using RINEX/SNR observations from selected coastal GNSS stations. Reflected GNSS signal patterns were processed to estimate reflector height and retrieve sea-level variations. The resulting time series were smoothed, followed by tidal reconstruction and de-tiding to isolate residual sea-level anomalies associated with storm-surge response during Hurricane Ian.

GNSS-Trop analysis used daily .trop files to extract integrated water vapor (IWV) time series at each station. GPM/IMERG HDF5 precipitation data were processed by day of year and matched to GNSS station coordinates using coordinate adjustment and bilinear interpolation. The station-based precipitation time series were then compared with GNSS-derived IWV to evaluate the relationship between atmospheric moisture buildup and rainfall during the storm event.

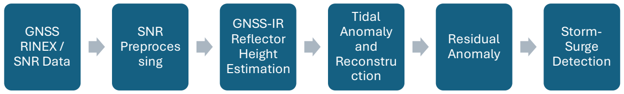


Figure 2. GNSS-IR Storm-Surge Detection Workflow. Processing workflow from GNSS RINEX/SNR data preprocessing to reflector height estimation, temporal smoothing, tidal reconstruction, residual anomaly extraction, and storm-surge detection.

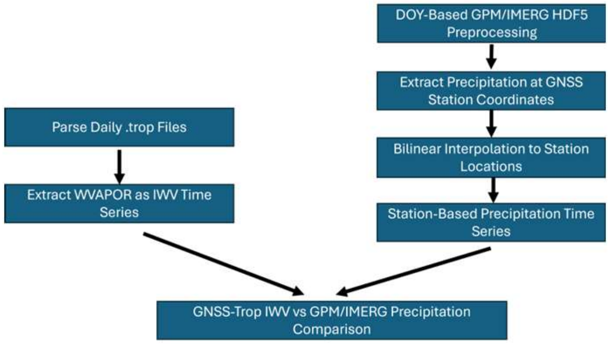
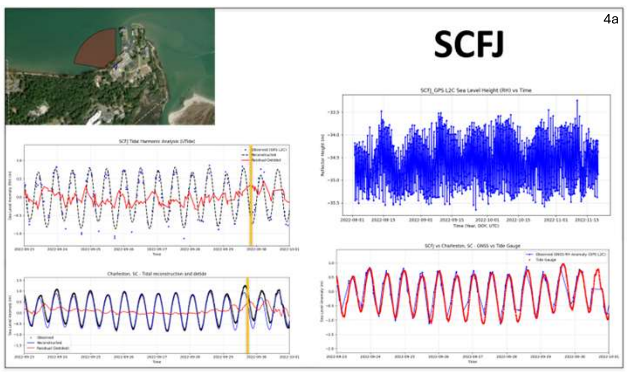
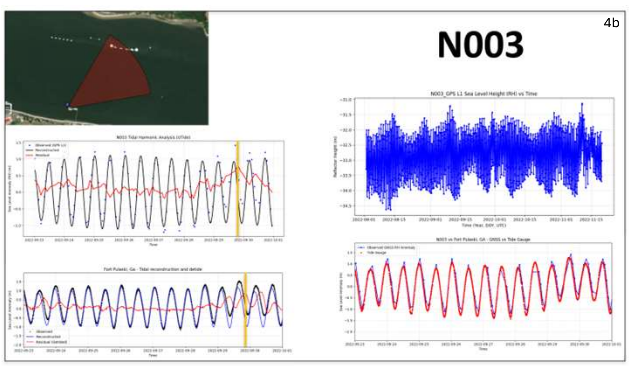


Figure 3. GNSS-Trop and IMERG Precipitation Comparison Workflow. Workflow for extracting GNSS-derived IWV and matching IMERG precipitation to GNSS station locations.

RESULTS





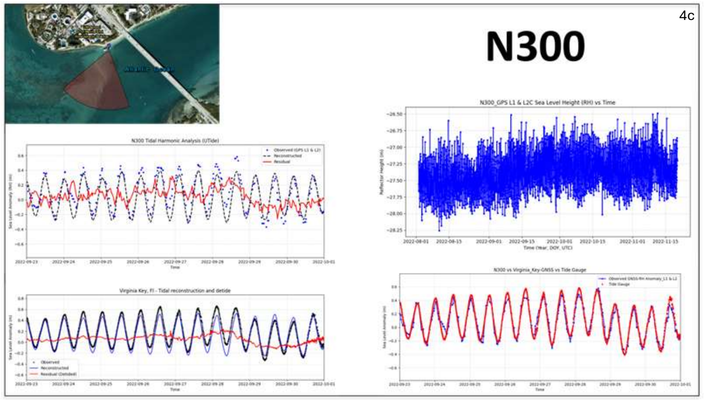


Figure 4. GNSS-IR Sea-Level Retrievals and Tide Gauge Comparison. Station-based GNSS-IR results for SCFJ (4a), N003 (4b), and N300 (4c) showing reflector height time series, UTide tidal reconstruction, de-tided residual anomalies, and comparison with nearby tide gauge observations during Hurricane Ian.

GNSS-IR retrievals showed clear tidal variability at all four stations, with reflector height time series capturing consistent sea-level oscillations during the Hurricane Ian event window. After UTide tidal reconstruction and de-tiding, residual anomalies highlighted non-tidal water-level changes associated with storm conditions. The strongest storm-related residual responses were observed at stations closer to the storm's main impact region, particularly SCFJ and N003 along the southeastern U.S. coast. Comparison with nearby tide gauges showed strong agreement between GNSS-derived sea-level anomalies and tide gauge observations, supporting the use of GNSS-IR as a complementary technique for coastal water-level and storm-surge monitoring.

GNSS-Trop integrated water vapor (IWV) showed clear temporal correlation with GPM/IMERG precipitation during Hurricane Ian, especially at stations along the southeastern United States. At SCFJ and N003, IWV increased before and during the main precipitation peaks, suggesting that GNSS-derived atmospheric moisture can capture storm-related moisture buildup and rainfall timing. N300 showed an earlier storm response consistent with its location along the storm track, indicating spatial variability as the hurricane moved northward.

The highlighted time windows show that elevated IWV often coincided with stronger IMERG rainfall bursts. In some cases, IWV began increasing before the heaviest precipitation, suggesting potential value for monitoring atmospheric conditions prior to intense rainfall. Overall, GNSS-Trop IWV provides useful atmospheric context for interpreting satellite precipitation during hurricane events.

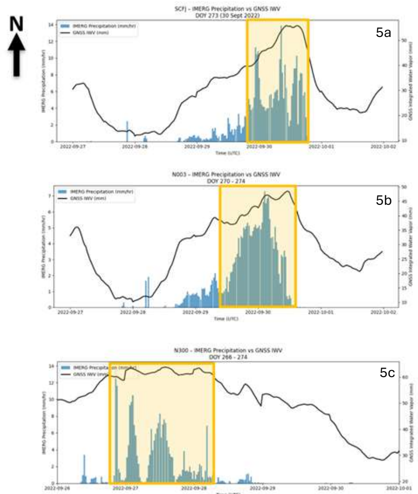


Figure 5. GNSS-Trop IWV and GPM/IMERG Precipitation. Comparison for **(5a) SCFJ**, **(5b) N003**, and **(5c) N300**, showing IWV changes and corresponding precipitation peaks during Hurricane Ian.

CONCLUSION

This study demonstrates that integrated GNSS observations can support hurricane coastal-hazard monitoring by capturing both oceanic and atmospheric storm responses. GNSS-IR successfully retrieved sea-level variations from reflected GNSS signals, while tidal reconstruction helped isolate residual non-tidal anomalies associated with storm surge. Comparison with co-located tide gauges showed strong agreement between the de-tided GNSS-IR sea-level anomalies and tide gauge observations. GNSS-Trop captured integrated water vapor evolution during Hurricane Ian, with IWV increases showing temporal correspondence with GPM/IMERG precipitation peaks, especially at stations closer to the storm's main impact region. Overall, combining GNSS-IR, GNSS-Trop, tide gauges, and satellite precipitation provides a resilient, all-weather framework for monitoring storm surge, atmospheric moisture, and rainfall during hurricane events.